5.9 EFM32WG380 (LQFP100)

5.9.1 Pinout

The EFM32WG380 pinout is shown in the following figure and table. Alternate locations are denoted by "#" followed by the location number (Multiple locations on the same pin are split with "/"). Alternate locations can be configured in the LOCATION bitfield in the *_ROUTE register in the module in question.

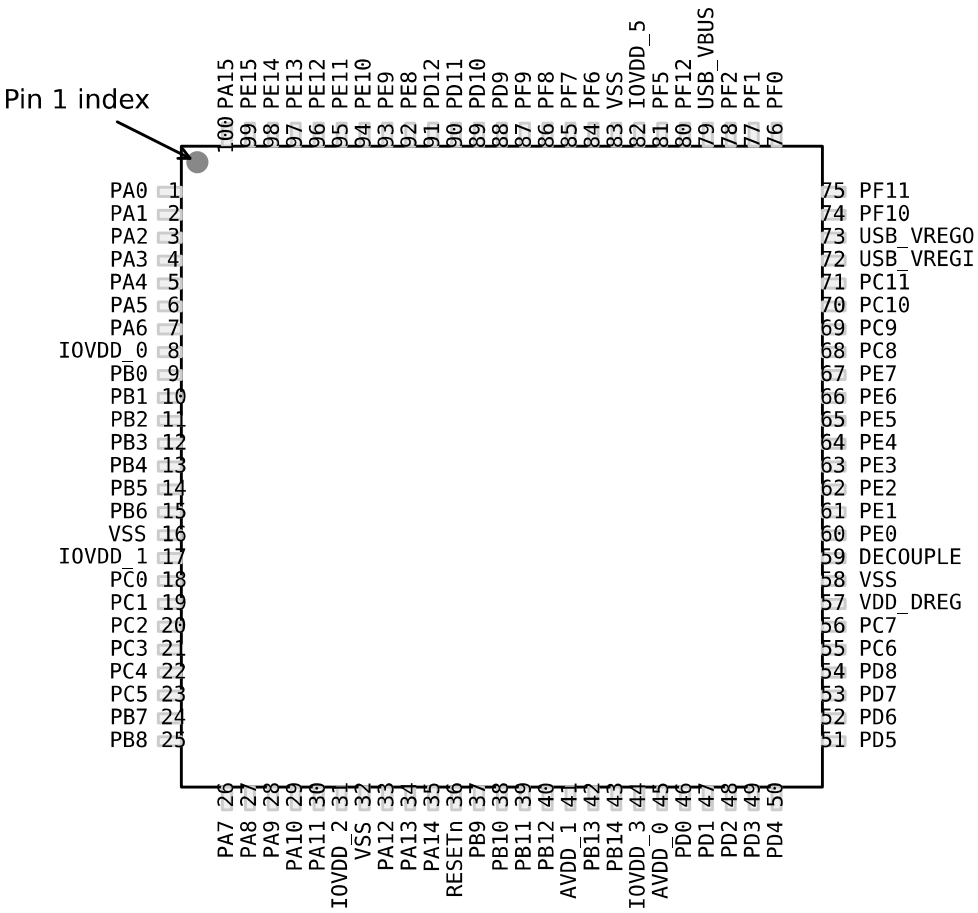


Figure 5.17. EFM32WG380 Pinout (top view, not to scale)

Table 5.25. Device Pinout

LQFP100 Pin# and Name		Pin Alternate Functionality / Description				
Pin #	Pin Name	Analog	EBI	Timers	Communication	Other
1	PA0		EBI_AD09 #0/1/2	TIM0_CC0 #0/1/4	LEU0_RX #4 I2C0_SDA #0	PRS_CH0 #0 GPIO_EM4WU0
2	PA1		EBI_AD10 #0/1/2	TIM0_CC1 #0/1	I2C0_SCL #0	CMU_CLK1 #0 PRS_CH1 #0
3	PA2		EBI_AD11 #0/1/2	TIM0_CC2 #0/1		CMU_CLK0 #0 ETM_TD0 #3
4	PA3		EBI_AD12 #0/1/2	TIM0_CDTI0 #0	U0_TX #2	LES_ALTEX2 #0 ETM_TD1 #3
5	PA4		EBI_AD13 #0/1/2	TIM0_CDTI1 #0	U0_RX #2	LES_ALTEX3 #0 ETM_TD2 #3
6	PA5		EBI_AD14 #0/1/2	TIM0_CDTI2 #0	LEU1_TX #1	LES_ALTEX4 #0 ETM_TD3 #3
7	PA6		EBI_AD15 #0/1/2		LEU1_RX #1	ETM_TCLK #3 GPIO_EM4WU1
8	IOVDD_0	Digital IO power supply 0.				
9	PB0		EBI_A16 #0/1/2	TIM1_CC0 #2		
10	PB1		EBI_A17 #0/1/2	TIM1_CC1 #2		
11	PB2		EBI_A18 #0/1/2	TIM1_CC2 #2		
12	PB3		EBI_A19 #0/1/2	PCNT1_S0IN #1	US2_TX #1	
13	PB4		EBI_A20 #0/1/2	PCNT1_S1IN #1	US2_RX #1	
14	PB5		EBI_A21 #0/1/2		US2_CLK #1	
15	PB6		EBI_A22 #0/1/2		US2_CS #1	
16	VSS	Ground.				
17	IOVDD_1	Digital IO power supply 1.				
18	PC0	ACMP0_CH0 DAC0_OUT0ALT #0/ OPAMP_OUT0ALT	EBI_A23 #0/1/2	TIM0_CC1 #4 PCNT0_S0IN #2	US0_TX #5 US1_TX #0 I2C0_SDA #4	LES_CH0 #0 PRS_CH2 #0
19	PC1	ACMP0_CH1 DAC0_OUT0ALT #1/ OPAMP_OUT0ALT	EBI_A24 #0/1/2	TIM0_CC2 #4 PCNT0_S1IN #2	US0_RX #5 US1_RX #0 I2C0_SCL #4	LES_CH1 #0 PRS_CH3 #0
20	PC2	ACMP0_CH2 DAC0_OUT0ALT #2/ OPAMP_OUT0ALT	EBI_A25 #0/1/2	TIM0_CDTI0 #4	US2_TX #0	LES_CH2 #0
21	PC3	ACMP0_CH3 DAC0_OUT0ALT #3/ OPAMP_OUT0ALT	EBI_NANDREn #0/1/2	TIM0_CDTI1 #4	US2_RX #0	LES_CH3 #0

LQFP100 Pin# and Name		Pin Alternate Functionality / Description				
Pin #	Pin Name	Analog	EBI	Timers	Communication	Other
22	PC4	ACMP0_CH4 / DAC0_P0 / OPAMP_P0	EBI_A26 #0/1/2	TIM0_CDTI2 #4 LE-TIM0_OUT0 #3 PCNT1_S0IN #0	US2_CLK #0 I2C1_SDA #0	LES_CH4 #0
23	PC5	ACMP0_CH5 / DAC0_N0 / OPAMP_N0	EBI_NANDWEn #0/1/2	LETIM0_OUT1 #3 PCNT1_S1IN #0	US2_CS #0 I2C1_SCL #0	LES_CH5 #0
24	PB7	LFXTAL_P		TIM1_CC0 #3	US0_TX #4 US1_CLK #0	
25	PB8	LFXTAL_N		TIM1_CC1 #3	US0_RX #4 US1_CS #0	
26	PA7		EBI_CSTFT #0/1/2			
27	PA8		EBI_DCLK #0/1/2	TIM2_CC0 #0		
28	PA9		EBI_DTEN #0/1/2	TIM2_CC1 #0		
29	PA10		EBI_VSNC #0/1/2	TIM2_CC2 #0		
30	PA11		EBI_HSNC #0/1/2			
31	IOVDD_2	Digital IO power supply 2.				
32	VSS	Ground.				
33	PA12		EBI_A00 #0/1/2	TIM2_CC0 #1		
34	PA13		EBI_A01 #0/1/2	TIM2_CC1 #1		
35	PA14		EBI_A02 #0/1/2	TIM2_CC2 #1		
36	RESETn	Reset input, active low. To apply an external reset source to this pin, it is required to only drive this pin low during reset, and let the internal pull-up ensure that reset is released.				
37	PB9		EBI_A03 #0/1/2		U1_TX #2	
38	PB10		EBI_A04 #0/1/2		U1_RX #2	
39	PB11	DAC0_OUT0 / OPAMP_OUT0		TIM1_CC2 #3 LE-TIM0_OUT0 #1	I2C1_SDA #1	
40	PB12	DAC0_OUT1 / OPAMP_OUT1		LETIM0_OUT1 #1	I2C1_SCL #1	
41	AVDD_1	Analog power supply 1.				
42	PB13	HFXTAL_P			US0_CLK #4/5 LEU0_TX #1	
43	PB14	HFXTAL_N			US0_CS #4/5 LEU0_RX #1	
44	IOVDD_3	Digital IO power supply 3.				
45	AVDD_0	Analog power supply 0.				
46	PD0	ADC0_CH0 DAC0_OUT0ALT #4/ OPAMP_OUT0ALT OPAMP_OUT2 #1		PCNT2_S0IN #0	US1_TX #1	

LQFP100 Pin# and Name		Pin Alternate Functionality / Description				
Pin #	Pin Name	Analog	EBI	Timers	Communication	Other
47	PD1	ADC0_CH1 DAC0_OUT1ALT #4/ OPAMP_OUT1ALT		TIM0_CC0 #3 PCNT2_S1IN #0	US1_RX #1	DBG_SWO #2
48	PD2	ADC0_CH2	EBI_A27 #0/1/2	TIM0_CC1 #3	USB_DMPU #0 US1_CLK #1	DBG_SWO #3
49	PD3	ADC0_CH3 OPAMP_N2		TIM0_CC2 #3	US1_CS #1	ETM_TD1 #0/2
50	PD4	ADC0_CH4 OPAMP_P2			LEU0_TX #0	ETM_TD2 #0/2
51	PD5	ADC0_CH5 OPAMP_OUT2 #0			LEU0_RX #0	ETM_TD3 #0/2
52	PD6	ADC0_CH6 / DAC0_P1 / OPAMP_P1		TIM1_CC0 #4 LE-TIM0_OUT0 #0 PCNT0_S0IN #3	US1_RX #2 I2C0_SDA #1	LES_ALTEX0 #0 ACMP0_O #2 ETM_TD0 #0
53	PD7	ADC0_CH7 / DAC0_N1 / OPAMP_N1		TIM1_CC1 #4 LE-TIM0_OUT1 #0 PCNT0_S1IN #3	US1_TX #2 I2C0_SCL #1	CMU_CLK0 #2 LES_ALTEX1 #0 ACMP1_O #2 ETM_TCLK #0
54	PD8	BU_VIN				CMU_CLK1 #1
55	PC6	ACMP0_CH6	EBI_A05 #0/1/2		LEU1_TX #0 I2C0_SDA #2	LES_CH6 #0 ETM_TCLK #2
56	PC7	ACMP0_CH7	EBI_A06 #0/1/2		LEU1_RX #0 I2C0_SCL #2	LES_CH7 #0 ETM_TD0 #2
57	VDD_DREG	Power supply for on-chip voltage regulator.				
58	VSS	Ground.				
59	DECOUPLE	Decouple output for on-chip voltage regulator. An external capacitance of size $C_{DECOUPLE}$ is required at this pin.				
60	PE0		EBI_A07 #0/1/2	TIM3_CC0 #1 PCNT0_S0IN #1	U0_TX #1 I2C1_SDA #2	
61	PE1		EBI_A08 #0/1/2	TIM3_CC1 #1 PCNT0_S1IN #1	U0_RX #1 I2C1_SCL #2	
62	PE2	BU_VOUT	EBI_A09 #0	TIM3_CC2 #1	U1_TX #3	ACMP0_O #1
63	PE3	BU_STAT	EBI_A10 #0		U1_RX #3	ACMP1_O #1
64	PE4		EBI_A11 #0/1/2		US0_CS #1	
65	PE5		EBI_A12 #0/1/2		US0_CLK #1	
66	PE6		EBI_A13 #0/1/2		US0_RX #1	
67	PE7		EBI_A14 #0/1/2		US0_TX #1	
68	PC8	ACMP1_CH0	EBI_A15 #0/1/2	TIM2_CC0 #2	US0_CS #2	LES_CH8 #0
69	PC9	ACMP1_CH1	EBI_A09 #1/2	TIM2_CC1 #2	US0_CLK #2	LES_CH9 #0 GPIO_EM4WU2
70	PC10	ACMP1_CH2	EBI_A10 #1/2	TIM2_CC2 #2	US0_RX #2	LES_CH10 #0
71	PC11	ACMP1_CH3	EBI_ALE #1/2		US0_TX #2	LES_CH11 #0

LQFP100 Pin# and Name		Pin Alternate Functionality / Description				
Pin #	Pin Name	Analog	EBI	Timers	Communication	Other
72	USB_VREGI	Input to internal 3.3V USB regulator				
73	USB_VREGO	Output and decoupling for internal 3.3V USB regulator				
74	PF10				U1_TX #1 USB_DM	
75	PF11				U1_RX #1 USB_DP	
76	PF0			TIM0_CC0 #5 LE-TIM0_OUT0 #2	US1_CLK #2 LEU0_TX #3 I2C0_SDA #5	DBG_SWCLK #0/1/2/3
77	PF1			TIM0_CC1 #5 LE-TIM0_OUT1 #2	US1_CS #2 LEU0_RX #3 I2C0_SCL #5	DBG_SWDIO #0/1/2/3 GPIO_EM4WU3
78	PF2		EBI_ARDY #0/1/2	TIM0_CC2 #5	LEU0_TX #4	ACMP1_O #0 DBG_SWO #0 GPIO_EM4WU4
79	USB_VBUS	USB 5.0 V VBUS input.				
80	PF12				USB_ID	
81	PF5		EBI_REn #0/2	TIM0_CDTI2 #2/5	USB_VBUSEN #0	PRS_CH2 #1
82	IOVDD_5	Digital IO power supply 5.				
83	VSS	Ground.				
84	PF6		EBI_BL0 #0/1/2	TIM0_CC0 #2	U0_TX #0	
85	PF7		EBI_BL1 #0/1/2	TIM0_CC1 #2	U0_RX #0	
86	PF8		EBI_WEn #1	TIM0_CC2 #2		ETM_TCLK #1
87	PF9		EBI_REn #1			ETM_TD0 #1
88	PD9		EBI_CS0 #0/1/2			
89	PD10		EBI_CS1 #0/1/2			
90	PD11		EBI_CS2 #0/1/2			
91	PD12		EBI_CS3 #0/1/2			
92	PE8		EBI_AD00 #0/1/2	PCNT2_S0IN #1		PRS_CH3 #1
93	PE9		EBI_AD01 #0/1/2	PCNT2_S1IN #1		
94	PE10		EBI_AD02 #0/1/2	TIM1_CC0 #1	US0_TX #0	BOOT_TX
95	PE11		EBI_AD03 #0/1/2	TIM1_CC1 #1	US0_RX #0	LES_ALTEX5 #0 BOOT_RX
96	PE12		EBI_AD04 #0/1/2	TIM1_CC2 #1	US0_RX #3 US0_CLK #0 I2C0_SDA #6	CMU_CLK1 #2 LES_ALTEX6 #0
97	PE13		EBI_AD05 #0/1/2		US0_TX #3 US0_CS #0 I2C0_SCL #6	LES_ALTEX7 #0 ACMP0_O #0 GPIO_EM4WU5
98	PE14		EBI_AD06 #0/1/2	TIM3_CC0 #0	LEU0_TX #2	
99	PE15		EBI_AD07 #0/1/2	TIM3_CC1 #0	LEU0_RX #2	
100	PA15		EBI_AD08 #0/1/2	TIM3_CC2 #0		

5.9.2 Alternate Functionality Pinout

A wide selection of alternate functionality is available for multiplexing to various pins. This is shown in the following table. The table shows the name of the alternate functionality in the first column, followed by columns showing the possible LOCATION bitfield settings.

Note: Some functionality, such as analog interfaces, do not have alternate settings or a LOCATION bitfield. In these cases, the pinout is shown in the column corresponding to LOCATION 0.

Table 5.26. Alternate Functionality Overview

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
ACMP0_CH0	PC0							Analog comparator ACMP0, channel 0.
ACMP0_CH1	PC1							Analog comparator ACMP0, channel 1.
ACMP0_CH2	PC2							Analog comparator ACMP0, channel 2.
ACMP0_CH3	PC3							Analog comparator ACMP0, channel 3.
ACMP0_CH4	PC4							Analog comparator ACMP0, channel 4.
ACMP0_CH5	PC5							Analog comparator ACMP0, channel 5.
ACMP0_CH6	PC6							Analog comparator ACMP0, channel 6.
ACMP0_CH7	PC7							Analog comparator ACMP0, channel 7.
ACMP0_O	PE13	PE2	PD6					Analog comparator ACMP0, digital output.
ACMP1_CH0	PC8							Analog comparator ACMP1, channel 0.
ACMP1_CH1	PC9							Analog comparator ACMP1, channel 1.
ACMP1_CH2	PC10							Analog comparator ACMP1, channel 2.
ACMP1_CH3	PC11							Analog comparator ACMP1, channel 3.
ACMP1_O	PF2	PE3	PD7					Analog comparator ACMP1, digital output.
ADC0_CH0	PD0							Analog to digital converter ADC0, input channel number 0.
ADC0_CH1	PD1							Analog to digital converter ADC0, input channel number 1.
ADC0_CH2	PD2							Analog to digital converter ADC0, input channel number 2.
ADC0_CH3	PD3							Analog to digital converter ADC0, input channel number 3.
ADC0_CH4	PD4							Analog to digital converter ADC0, input channel number 4.
ADC0_CH5	PD5							Analog to digital converter ADC0, input channel number 5.
ADC0_CH6	PD6							Analog to digital converter ADC0, input channel number 6.
ADC0_CH7	PD7							Analog to digital converter ADC0, input channel number 7.
BOOT_RX	PE11							Bootloader RX.
BOOT_TX	PE10							Bootloader TX.

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
BU_STAT	PE3							Backup Power Domain status, whether or not the system is in backup mode
BU_VIN	PD8							Battery input for Backup Power Domain
BU_VOUT	PE2							Power output for Backup Power Domain
CMU_CLK0	PA2		PD7					Clock Management Unit, clock output number 0.
CMU_CLK1	PA1	PD8	PE12					Clock Management Unit, clock output number 1.
DAC0_OUT0 / OPAMP_OUT0	PB11							Digital to Analog Converter DAC0_OUT0 /OPAMP output channel number 0.
DAC0_OUT0ALT / OPAMP_OUT0ALT	PC0	PC1	PC2	PC3	PD0			Digital to Analog Converter DAC0_OUT0ALT / OPAMP alternative output for channel 0.
DAC0_OUT1 / OPAMP_OUT1	PB12							Digital to Analog Converter DAC0_OUT1 /OPAMP output channel number 1.
DAC0_OUT1ALT / OPAMP_OUT1ALT					PD1			Digital to Analog Converter DAC0_OUT1ALT / OPAMP alternative output for channel 1.
DBG_SWCLK	PF0	PF0	PF0	PF0				Debug-interface Serial Wire clock input. Note that this function is enabled to pin out of reset, and has a built-in pull down.
DBG_SWDIO	PF1	PF1	PF1	PF1				Debug-interface Serial Wire data input / output. Note that this function is enabled to pin out of reset, and has a built-in pull up.
DBG_SWO	PF2		PD1	PD2				Debug-interface Serial Wire viewer Output. Note that this function is not enabled after reset, and must be enabled by software to be used.
EBI_A00	PA12	PA12	PA12					External Bus Interface (EBI) address output pin 00.
EBI_A01	PA13	PA13	PA13					External Bus Interface (EBI) address output pin 01.
EBI_A02	PA14	PA14	PA14					External Bus Interface (EBI) address output pin 02.
EBI_A03	PB9	PB9	PB9					External Bus Interface (EBI) address output pin 03.
EBI_A04	PB10	PB10	PB10					External Bus Interface (EBI) address output pin 04.
EBI_A05	PC6	PC6	PC6					External Bus Interface (EBI) address output pin 05.
EBI_A06	PC7	PC7	PC7					External Bus Interface (EBI) address output pin 06.
EBI_A07	PE0	PE0	PE0					External Bus Interface (EBI) address output pin 07.
EBI_A08	PE1	PE1	PE1					External Bus Interface (EBI) address output pin 08.
EBI_A09	PE2	PC9	PC9					External Bus Interface (EBI) address output pin 09.
EBI_A10	PE3	PC10	PC10					External Bus Interface (EBI) address output pin 10.
EBI_A11	PE4	PE4	PE4					External Bus Interface (EBI) address output pin 11.
EBI_A12	PE5	PE5	PE5					External Bus Interface (EBI) address output pin 12.
EBI_A13	PE6	PE6	PE6					External Bus Interface (EBI) address output pin 13.
EBI_A14	PE7	PE7	PE7					External Bus Interface (EBI) address output pin 14.

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
EBI_A15	PC8	PC8	PC8					External Bus Interface (EBI) address output pin 15.
EBI_A16	PB0	PB0	PB0					External Bus Interface (EBI) address output pin 16.
EBI_A17	PB1	PB1	PB1					External Bus Interface (EBI) address output pin 17.
EBI_A18	PB2	PB2	PB2					External Bus Interface (EBI) address output pin 18.
EBI_A19	PB3	PB3	PB3					External Bus Interface (EBI) address output pin 19.
EBI_A20	PB4	PB4	PB4					External Bus Interface (EBI) address output pin 20.
EBI_A21	PB5	PB5	PB5					External Bus Interface (EBI) address output pin 21.
EBI_A22	PB6	PB6	PB6					External Bus Interface (EBI) address output pin 22.
EBI_A23	PC0	PC0	PC0					External Bus Interface (EBI) address output pin 23.
EBI_A24	PC1	PC1	PC1					External Bus Interface (EBI) address output pin 24.
EBI_A25	PC2	PC2	PC2					External Bus Interface (EBI) address output pin 25.
EBI_A26	PC4	PC4	PC4					External Bus Interface (EBI) address output pin 26.
EBI_A27	PD2	PD2	PD2					External Bus Interface (EBI) address output pin 27.
EBI_AD00	PE8	PE8	PE8					External Bus Interface (EBI) address and data input / output pin 00.
EBI_AD01	PE9	PE9	PE9					External Bus Interface (EBI) address and data input / output pin 01.
EBI_AD02	PE10	PE10	PE10					External Bus Interface (EBI) address and data input / output pin 02.
EBI_AD03	PE11	PE11	PE11					External Bus Interface (EBI) address and data input / output pin 03.
EBI_AD04	PE12	PE12	PE12					External Bus Interface (EBI) address and data input / output pin 04.
EBI_AD05	PE13	PE13	PE13					External Bus Interface (EBI) address and data input / output pin 05.
EBI_AD06	PE14	PE14	PE14					External Bus Interface (EBI) address and data input / output pin 06.
EBI_AD07	PE15	PE15	PE15					External Bus Interface (EBI) address and data input / output pin 07.
EBI_AD08	PA15	PA15	PA15					External Bus Interface (EBI) address and data input / output pin 08.
EBI_AD09	PA0	PA0	PA0					External Bus Interface (EBI) address and data input / output pin 09.
EBI_AD10	PA1	PA1	PA1					External Bus Interface (EBI) address and data input / output pin 10.
EBI_AD11	PA2	PA2	PA2					External Bus Interface (EBI) address and data input / output pin 11.
EBI_AD12	PA3	PA3	PA3					External Bus Interface (EBI) address and data input / output pin 12.
EBI_AD13	PA4	PA4	PA4					External Bus Interface (EBI) address and data input / output pin 13.

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
EBI_AD14	PA5	PA5	PA5					External Bus Interface (EBI) address and data input / output pin 14.
EBI_AD15	PA6	PA6	PA6					External Bus Interface (EBI) address and data input / output pin 15.
EBI_ALE		PC11	PC11					External Bus Interface (EBI) Address Latch Enable output.
EBI_ARDY	PF2	PF2	PF2					External Bus Interface (EBI) Hardware Ready Control input.
EBI_BL0	PF6	PF6	PF6					External Bus Interface (EBI) Byte Lane/Enable pin 0.
EBI_BL1	PF7	PF7	PF7					External Bus Interface (EBI) Byte Lane/Enable pin 1.
EBI_CS0	PD9	PD9	PD9					External Bus Interface (EBI) Chip Select output 0.
EBI_CS1	PD10	PD10	PD10					External Bus Interface (EBI) Chip Select output 1.
EBI_CS2	PD11	PD11	PD11					External Bus Interface (EBI) Chip Select output 2.
EBI_CS3	PD12	PD12	PD12					External Bus Interface (EBI) Chip Select output 3.
EBI_CSTFT	PA7	PA7	PA7					External Bus Interface (EBI) Chip Select output TFT.
EBI_DCLK	PA8	PA8	PA8					External Bus Interface (EBI) TFT Dot Clock pin.
EBI_DTEN	PA9	PA9	PA9					External Bus Interface (EBI) TFT Data Enable pin.
EBI_HSNC	PA11	PA11	PA11					External Bus Interface (EBI) TFT Horizontal Synchronization pin.
EBI_NANDREn	PC3	PC3	PC3					External Bus Interface (EBI) NAND Read Enable output.
EBI_NANDWEn	PC5	PC5	PC5					External Bus Interface (EBI) NAND Write Enable output.
EBI_REn	PF5	PF9	PF5					External Bus Interface (EBI) Read Enable output.
EBI_VSNC	PA10	PA10	PA10					External Bus Interface (EBI) TFT Vertical Synchronization pin.
EBI_WEn		PF8						External Bus Interface (EBI) Write Enable output.
ETM_TCLK	PD7	PF8	PC6	PA6				Embedded Trace Module ETM clock .
ETM_TD0	PD6	PF9	PC7	PA2				Embedded Trace Module ETM data 0.
ETM_TD1	PD3		PD3	PA3				Embedded Trace Module ETM data 1.
ETM_TD2	PD4		PD4	PA4				Embedded Trace Module ETM data 2.
ETM_TD3	PD5		PD5	PA5				Embedded Trace Module ETM data 3.
GPIO_EM4WU0	PA0							Pin can be used to wake the system up from EM4
GPIO_EM4WU1	PA6							Pin can be used to wake the system up from EM4
GPIO_EM4WU2	PC9							Pin can be used to wake the system up from EM4
GPIO_EM4WU3	PF1							Pin can be used to wake the system up from EM4
GPIO_EM4WU4	PF2							Pin can be used to wake the system up from EM4
GPIO_EM4WU5	PE13							Pin can be used to wake the system up from EM4

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
HFXTAL_N	PB14							High Frequency Crystal negative pin. Also used as external optional clock input pin.
HFXTAL_P	PB13							High Frequency Crystal positive pin.
I2C0_SCL	PA1	PD7	PC7		PC1	PF1	PE13	I2C0 Serial Clock Line input / output.
I2C0_SDA	PA0	PD6	PC6		PC0	PF0	PE12	I2C0 Serial Data input / output.
I2C1_SCL	PC5	PB12	PE1					I2C1 Serial Clock Line input / output.
I2C1_SDA	PC4	PB11	PE0					I2C1 Serial Data input / output.
LES_ALTEX0	PD6							LESENSE alternate exite output 0.
LES_ALTEX1	PD7							LESENSE alternate exite output 1.
LES_ALTEX2	PA3							LESENSE alternate exite output 2.
LES_ALTEX3	PA4							LESENSE alternate exite output 3.
LES_ALTEX4	PA5							LESENSE alternate exite output 4.
LES_ALTEX5	PE11							LESENSE alternate exite output 5.
LES_ALTEX6	PE12							LESENSE alternate exite output 6.
LES_ALTEX7	PE13							LESENSE alternate exite output 7.
LES_CH0	PC0							LESENSE channel 0.
LES_CH1	PC1							LESENSE channel 1.
LES_CH2	PC2							LESENSE channel 2.
LES_CH3	PC3							LESENSE channel 3.
LES_CH4	PC4							LESENSE channel 4.
LES_CH5	PC5							LESENSE channel 5.
LES_CH6	PC6							LESENSE channel 6.
LES_CH7	PC7							LESENSE channel 7.
LES_CH8	PC8							LESENSE channel 8.
LES_CH9	PC9							LESENSE channel 9.
LES_CH10	PC10							LESENSE channel 10.
LES_CH11	PC11							LESENSE channel 11.
LETIM0_OUT0	PD6	PB11	PF0	PC4				Low Energy Timer LETIM0, output channel 0.
LETIM0_OUT1	PD7	PB12	PF1	PC5				Low Energy Timer LETIM0, output channel 1.
LEU0_RX	PD5	PB14	PE15	PF1	PA0			LEUART0 Receive input.
LEU0_TX	PD4	PB13	PE14	PF0	PF2			LEUART0 Transmit output. Also used as receive input in half duplex communication.
LEU1_RX	PC7	PA6						LEUART1 Receive input.
LEU1_TX	PC6	PA5						LEUART1 Transmit output. Also used as receive input in half duplex communication.
LFXTAL_N	PB8							Low Frequency Crystal (typically 32.768 kHz) negative pin. Also used as an optional external clock input pin.

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
LFXTAL_P	PB7							Low Frequency Crystal (typically 32.768 kHz) positive pin.
OPAMP_N0 / DAC0_N0	PC5							Operational Amplifier 0 external negative input.
OPAMP_N1 / DAC0_N1	PD7							Operational Amplifier 1 external negative input.
OPAMP_N2	PD3							Operational Amplifier 2 external negative input.
OPAMP_OUT2	PD5	PD0						Operational Amplifier 2 output.
OPAMP_P0 / DAC0_P0	PC4							Operational Amplifier 0 external positive input.
OPAMP_P1 / DAC0_P1	PD6							Operational Amplifier 1 external positive input.
OPAMP_P2	PD4							Operational Amplifier 2 external positive input.
PCNT0_S0IN		PE0	PC0	PD6				Pulse Counter PCNT0 input number 0.
PCNT0_S1IN		PE1	PC1	PD7				Pulse Counter PCNT0 input number 1.
PCNT1_S0IN	PC4	PB3						Pulse Counter PCNT1 input number 0.
PCNT1_S1IN	PC5	PB4						Pulse Counter PCNT1 input number 1.
PCNT2_S0IN	PD0	PE8						Pulse Counter PCNT2 input number 0.
PCNT2_S1IN	PD1	PE9						Pulse Counter PCNT2 input number 1.
PRS_CH0	PA0							Peripheral Reflex System PRS, channel 0.
PRS_CH1	PA1							Peripheral Reflex System PRS, channel 1.
PRS_CH2	PC0	PF5						Peripheral Reflex System PRS, channel 2.
PRS_CH3	PC1	PE8						Peripheral Reflex System PRS, channel 3.
TIM0_CC0	PA0	PA0	PF6	PD1	PA0	PF0		Timer 0 Capture Compare input / output channel 0.
TIM0_CC1	PA1	PA1	PF7	PD2	PC0	PF1		Timer 0 Capture Compare input / output channel 1.
TIM0_CC2	PA2	PA2	PF8	PD3	PC1	PF2		Timer 0 Capture Compare input / output channel 2.
TIM0_CDTI0	PA3				PC2			Timer 0 Complimentary Dead Time Insertion channel 0.
TIM0_CDTI1	PA4				PC3			Timer 0 Complimentary Dead Time Insertion channel 1.
TIM0_CDTI2	PA5		PF5		PC4	PF5		Timer 0 Complimentary Dead Time Insertion channel 2.
TIM1_CC0		PE10	PB0	PB7	PD6			Timer 1 Capture Compare input / output channel 0.
TIM1_CC1		PE11	PB1	PB8	PD7			Timer 1 Capture Compare input / output channel 1.
TIM1_CC2		PE12	PB2	PB11				Timer 1 Capture Compare input / output channel 2.
TIM2_CC0	PA8	PA12	PC8					Timer 2 Capture Compare input / output channel 0.
TIM2_CC1	PA9	PA13	PC9					Timer 2 Capture Compare input / output channel 1.
TIM2_CC2	PA10	PA14	PC10					Timer 2 Capture Compare input / output channel 2.
TIM3_CC0	PE14	PE0						Timer 3 Capture Compare input / output channel 0.

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
TIM3_CC1	PE15	PE1						Timer 3 Capture Compare input / output channel 1.
TIM3_CC2	PA15	PE2						Timer 3 Capture Compare input / output channel 2.
U0_RX	PF7	PE1	PA4					UART0 Receive input.
U0_TX	PF6	PE0	PA3					UART0 Transmit output. Also used as receive input in half duplex communication.
U1_RX		PF11	PB10	PE3				UART1 Receive input.
U1_TX		PF10	PB9	PE2				UART1 Transmit output. Also used as receive input in half duplex communication.
US0_CLK	PE12	PE5	PC9		PB13	PB13		USART0 clock input / output.
US0_CS	PE13	PE4	PC8		PB14	PB14		USART0 chip select input / output.
US0_RX	PE11	PE6	PC10	PE12	PB8	PC1		USART0 Asynchronous Receive. USART0 Synchronous mode Main Input / Secondary Output (MISO).
US0_TX	PE10	PE7	PC11	PE13	PB7	PC0		USART0 Asynchronous Transmit.Also used as receive input in half duplex communication. USART0 Synchronous mode Main Output / Secondary Input (MOSI).
US1_CLK	PB7	PD2	PF0					USART1 clock input / output.
US1_CS	PB8	PD3	PF1					USART1 chip select input / output.
US1_RX	PC1	PD1	PD6					USART1 Asynchronous Receive. USART1 Synchronous mode Main Input / Secondary Output (MISO).
US1_TX	PC0	PD0	PD7					USART1 Asynchronous Transmit.Also used as receive input in half duplex communication. USART1 Synchronous mode Main Output / Secondary Input (MOSI).
US2_CLK	PC4	PB5						USART2 clock input / output.
US2_CS	PC5	PB6						USART2 chip select input / output.
US2_RX	PC3	PB4						USART2 Asynchronous Receive. USART2 Synchronous mode Main Input / Secondary Output (MISO).
US2_TX	PC2	PB3						USART2 Asynchronous Transmit.Also used as receive input in half duplex communication. USART2 Synchronous mode Main Output / Secondary Input (MOSI).
USB_DM	PF10							USB D- pin.
USB_DMPU	PD2							USB D- Pullup control.
USB_DP	PF11							USB D+ pin.
USB_ID	PF12							USB ID pin. Used in OTG mode.
USB_VBUS	USB_VBUS							USB 5 V VBUS input.

Alternate	LOCATION							
Functionality	0	1	2	3	4	5	6	Description
USB_VBUSEN	PF5							USB 5 V VBUS enable.
USB_VREGI	USB_VREGI							USB Input to internal 3.3 V regulator
USB_VREGO	USB_VREGO							USB Decoupling for internal 3.3 V USB regulator and regulator output

5.9.3 GPIO Pinout Overview

The specific GPIO pins available in EFM32WG380 are shown in the following table. Each GPIO port is organized as 16-bit ports indicated by letters A through F, and the individual pin on this port is indicated by a number from 15 down to 0.

Table 5.27. GPIO Pinout

Port	Pin 15	Pin 14	Pin 13	Pin 12	Pin 11	Pin 10	Pin 9	Pin 8	Pin 7	Pin 6	Pin 5	Pin 4	Pin 3	Pin 2	Pin 1	Pin 0
Port A	PA15	PA14	PA13	PA12	PA11	PA10	PA9	PA8	PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0
Port B	—	PB14	PB13	PB12	PB11	PB10	PB9	PB8	PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
Port C	—	—	—	—	PC11	PC10	PC9	PC8	PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
Port D	—	—	—	PD12	PD11	PD10	PD9	PD8	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0
Port E	PE15	PE14	PE13	PE12	PE11	PE10	PE9	PE8	PE7	PE6	PE5	PE4	PE3	PE2	PE1	PE0
Port F	—	—	—	PF12	PF11	PF10	PF9	PF8	PF7	PF6	PF5	—	—	PF2	PF1	PF0

5.9.4 Opamp Pinout Overview

The specific opamp terminals available in EFM32WG380 is shown in the following figure.

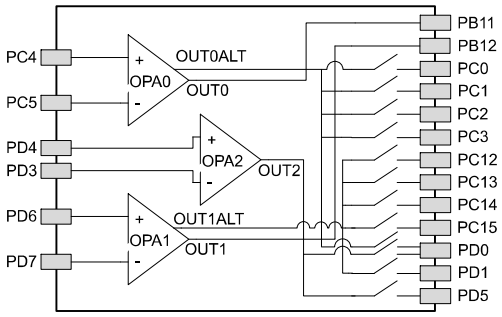


Figure 5.18. Opamp Pinout